

# Linear equations

While TSM 099 studies Topic 27. Review: Topic 24, Topic 25, Topic 26.

Equivalent forms are acceptable. Use these answers after attempting the problems; revisit the linked lecture notes for additional examples.

## Core: two-term and short linear equations

1. Solve and check in the original equation.

$$5x + 7 = 37$$

$$x = 6$$

Check by substituting into the original equation.

2. Solve and check in the original equation.

$$-3x + 1 = 7$$

$$x = -2$$

Check by substituting into the original equation.

3. Solve and check in the original equation.

$$-6x - 6 = 0$$

$$x = -1$$

Check by substituting into the original equation.

4. Solve and check in the original equation.

$$3x + 2 = -1$$

$$x = -1$$

Check by substituting into the original equation.

5. Solve and check in the original equation.

$$-3x + 4 = 13$$

$$x = -3$$

Check by substituting into the original equation.

6. Solve and check in the original equation.

$$-4x - 1 = -9$$

$$x = 2$$

Check by substituting into the original equation.

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## Core: variable terms on both sides

**7.** Solve and check in the original equation.

$$-5x + 2 = -2x + 5$$

$$x = -1$$

Check by substituting into the original equation.

**8.** Solve and check in the original equation.

$$4x + 3 = -2x + 6$$

$$x = \frac{1}{2}$$

Check by substituting into the original equation.

**9.** Solve and check in the original equation.

$$2x + 6 = -2x + 7$$

$$x = \frac{1}{4}$$

Check by substituting into the original equation.

**10.** Solve and check in the original equation.

$$2x + 2 = x + 2$$

$$x = 0$$

Check by substituting into the original equation.

**11.** Solve and check in the original equation.

$$-3x - 6 = x + 4$$

$$x = -\frac{5}{2}$$

Check by substituting into the original equation.

**12.** Solve and check in the original equation.

$$2x + 5 = -2x + 8$$

$$x = \frac{3}{4}$$

Check by substituting into the original equation.

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## Core: equations with parentheses

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**13.** Solve and check in the original equation.

$$-3x - 13 = 2x - 8$$

$$x = -1$$

Check by substituting into the original equation.

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**14.** Solve and check in the original equation.

$$-3x + 18 = 27$$

$$x = -3$$

Check by substituting into the original equation.

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**15.** Solve and check in the original equation.

$$6x + 2 = -3x + 20$$

$$x = 2$$

Check by substituting into the original equation.

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**16.** Solve and check in the original equation.

$$6x - 16 = -x - 23$$

$$x = -1$$

Check by substituting into the original equation.

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**17.** Solve and check in the original equation.

$$-4x - 8 = x - 23$$

$$x = 3$$

Check by substituting into the original equation.

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**18.** Solve and check in the original equation.

$$-2x + 7 = 3x - 3$$

$$x = 2$$

Check by substituting into the original equation.

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## Extension: mixed equations and special cases

**19.** Solve and check in the original equation.

$$5x - 3 = 2x + 8$$

$$x = \frac{11}{3}$$

Check by substituting into the original equation.

**20.** Solve and check in the original equation.

$$-4x + 9 = 3x - 2$$

$$x = \frac{11}{7}$$

Check by substituting into the original equation.

**21.** Solve and check in the original equation.

$$6x - 11 = 2x + 5$$

$$x = 4$$

Check by substituting into the original equation.

**22.** Solve and check in the original equation.

$$-6x - 1 = 3x - 7$$

$$x = \frac{2}{3}$$

Check by substituting into the original equation.

**23.** Solve and check in the original equation.

$$2x + 6 = 2x + 6$$

All real numbers (the two sides are identical).

**24.** Solve and check in the original equation.

$$3x - 6 = 3x + 1$$

No solution (the two sides cannot be equal).